ABSTRACT

Diabetic retinal diseases, including diabetic retinopathy and glaucoma, are irreversible causes of blindness, demanding early detection to mitigate vision loss. This study proposes a novel soft computing-based framework for the prediction of diabetic retinal diseases and glaucoma within diabetic populations, leveraging advanced computational techniques to enhance diagnostic accuracy.

This work introduces a multi-model architecture integrating four innovative soft computing approaches. Firstly, a Random Search Optimization Adaboost Ensemble classifier is designed for diabetic retinal disease and glaucoma prediction, combining adaptive boosting with hyperparameter optimization to address dataset imbalance and improve generalization. Secondly, a Deep Neural Classification Network is developed for diabetic retinal disease detection, employing Region of Interest guided segmentation and Neural Image Optimization, followed by Grey Level Co-occurrence Matrix feature extraction to reduce dimensionality and enhance lesion discrimination. Next, an Artificial Algae Algorithm with Support Vector Machine hybrid model is proposed for glaucoma diagnosis, integrating bio-inspired optimization for feature selection with kernel-based classification to improve boundary delineation in fundus images. Finally, an Interpretable Hybrid Stacking Ensemble Classifier leverages Explainable Artificial Intelligence principles to detect glaucoma in diabetic patients, combining Grey Co-occurrence Matrix and Grey Level Run Length Matrix textural features with meta-learning for transparent decision-making.

The framework is validated using retinal datasets, processed through multilevel preprocessing (noise removal, homomorphic filtering, color normalization) and segmented via Ternausnet and Faster Region-based Convolutional Neural Networks for structural feature extraction. Experimental results demonstrate that the proposed models outperform conventional methods, achieving superior precision 94.3%, recall 92.8%, F1 score 93.5%, and classification accuracy 96.1% with minimized error rates 3.2%. The Random Search Optimization Adaboost Ensemble and Explainable Artificial Intelligence Hybrid Stacking Ensemble Classification models exhibit exceptional interpretability, addressing the "black-box" limitation of deep learning, while the Artificial Algae Algorithm Support Vector

Machine and Deep Neural Classification Network optimize computational efficiency and specificity.

This study highlights the transformative potential of soft computing techniques ensemble learning, deep feature engineering, bio-inspired optimization, and explainable artificial intelligence in advancing early diagnosis of diabetic eye diseases. By harmonizing accuracy, interpretability, and clinical relevance, the proposed framework offers a scalable solution for reducing preventable blindness in diabetic communities, aligning with global healthcare priorities. This research underscores the transformative role of artificial intelligence in advancing health justice, ensuring that diabetic communities regardless of geography or socioeconomic status can access timely, life changing interventions